

Ariel Trejo

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WORK EXPERIENCE

Library of Congress, Federal Government, Remote

Software Engineer Intern

June 2026 – Aug 2026

- Developed and implemented 126 unit tests using xUnit and Moq for an enterprise ASP.NET Core (.NET 10) modernization project.
- Increased automated unit test coverage to 84%, achieving 1,176 covered code blocks while reducing uncovered blocks to 221.
- Validated repository, service, and data access layers by testing business logic, Entity Framework Core queries, dependency injection, and edge-case scenarios.
- Collaborated within an Agile/Scrum development team to support the migration of legacy .NET Framework applications to modern ASP.NET Core microservices.

EOS Additive Manufacturing, Pflugerville, TX

Returning Data Analysis Intern

May 2025 – Aug 2025

- Migrated the company's static HTML database interface into a full React.js dashboard with dynamic, real-time data interaction.
- Implemented interactive visualization pipelines using D3.js and Recharts; upgraded tables to MUI DataGrid for high-volume performance.
- Deployed the platform on an Azure Virtual Machine and configured firewall rules, access controls, and IP restrictions.

Data Analysis Intern

June 2024 – Aug 2024

- Built a full-stack data management system processing thousands of manufacturing records.
- Configured and connected a PostgreSQL database to a Python-based data pipeline, including data type correction and schema cleanup for reliable querying.
- Developed backend services using Python + Flask; built a responsive frontend in HTML, JavaScript, and Bootstrap.

UTRGV Institute for Advanced Manufacturing, Research Assistant, Edinburg, TX

Oct 2023 – Jan 2026

Digital Twin Research Project

- Developed a Digital Twin which consists of augmented reality using Microsoft HoloLens to produce a 3-D model of a KUKA AG which is interactable for the user, utilizing Unity, C# and Vuforia.
- Created an Azure SQL Database to connect with HoloLens, enabling users to interact with a digital map and retrieve object information.

EDUCATION

The University of Texas Rio Grande Valley, Edinburg, TX

B.S. in Computer Science, GPA: 3.77/4.0

Publications: [Enhancing Immersive Experiences: Computer Vision Enhanced Mixed Reality in Smart Manufacturing](#)

TECHNICAL SKILLS

Programming Languages: Python, C++, C, C#, SQL, PostgreSQL, JavaScript, HTML

Technology, Tools, & Frameworks: Entity Framework Core, xUnit, Moq, Azure DevOps, ASP.NET Core (.NET 10), React.js (React), Microsoft Azure (Azure), Azure Virtual Machines, SQLAlchemy, Unity, Flask Framework, LangChain, Ollama, D3.js, Recharts, JSON, Bootstrap, Network Security / Firewall Configuration, Large Language Models (LLMs), Generative AI, Digital Twin, Augmented Reality (AR), Robotics, Agile/Scrum